DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM
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ALGEBRA (Ph.D. Version)

Instructions to the student

a. Answer all six questions; each will be assigned a grade from 0 to 10.

b. Use a different booklet for each question. Write the problem number and your **code number (not** your name) on the outside of the booklet.

c. Keep scratch work on separate pages in the same booklet.

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1. Let p be a prime number and let G be a group of order p^n for some $n \geq 1$. Let G' be the commutator subgroup of G.

(a) Let H be a subgroup of G with $[G : H] = p$. Show that $G' \subseteq H$.

(b) Let

$$F_G = \bigcap_{[G:H]=p} H$$

be the intersection of all subgroups of G of index p. Show that F is a normal subgroup of G and that G/F is an elementary abelian p-group (an elementary abelian p-group A is an abelian group such that $a^p = 1$ for all $a \in A$).

(c) Let A be a finite elementary abelian p-group. Show that $F_A = 1$ (where F_A, as in (b), is the intersection of all subgroups of A of index p).

(d) Let B be a normal subgroup of G such that G/B is an elementary abelian p-group. Show that $F_G \subseteq B$.

2. Let m and n be positive integers. Let V be the set of $m \times n$ matrices over $\mathbb{C}$ and let W be the set of $n \times m$ matrices over $\mathbb{C}$. Both V and W are vector spaces of dimension mn over $\mathbb{C}$. Let $A \in V$. Consider the linear functional on W:

$$\phi_A : W \to \mathbb{C}, \quad \phi_A(B) = \text{Trace}(AB),$$

where the trace of a square matrix is the sum of the diagonal elements.
(a) Show that if $A \neq 0$ then ϕ_A is not the zero map.
(b) Show that every linear functional on W is of the form ϕ_A for some $A \in V$.

3. Let R be a commutative ring with 1. Let M be an R-module and let $0 \to A \to B \to C \to 0$ be an exact sequence of R-modules. It is known that

$$A \otimes M \to B \otimes M \to C \otimes M \to 0$$

is always exact (where $\otimes$ denotes $\otimes_R$). If $A \otimes M \to B \otimes M$ is an injection for all such exact sequences $0 \to A \to B \to C \to 0$, then M is said to be *flat*. Suppose that $0 \to M_1 \to M_2 \to M \to 0$ is an exact sequence of R-modules. Let N be an R-module. This problem will show that if M is flat then

$$0 \to M_1 \otimes N \to M_2 \otimes N \to M \otimes N \to 0$$

is exact. This only requires showing that $M_1 \otimes N \to M_2 \otimes N$ is an injection.
(a) Show that a free module is flat.
(b) Write $N = F/K$, where F is free. Use the diagram

$$\begin{tikzcd}
 & & & 0 \arrow[d] & \\
 & M_1 \otimes K \arrow[r] \arrow[d] & M_2 \otimes K \arrow[r] \arrow[d] & M \otimes K \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & M_1 \otimes F \arrow[r] \arrow[d] & M_2 \otimes F \arrow[r] \arrow[d] & M \otimes F \arrow[r] \arrow[d] & 0 \\
 & M_1 \otimes N \arrow[r] \arrow[d] & M_2 \otimes N \arrow[r] \arrow[d] & M \otimes N \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

to show that $M_1 \otimes N \to M_2 \otimes N$ is an injection when M is flat. (*Note:* It is also permissible to prove this part using properties of "Tor")

4. Let R be a principal ideal domain (PID).
(a) Show that all nonzero prime ideals of R are maximal ideals.
(b) Let S be an integral domain that is not a field. Suppose that there is a surjective ring homomorphism $\phi : R \to S$. Show that ϕ is an isomorphism. (*Note:* an integral domain is not allowed to be the zero ring.)

5. Let L/K be a finite Galois extension of fields and let $G = \mathrm{Gal}(L/K)$. Let $\alpha \in L$ with $\alpha \notin K$.
(a) Show that there exists $g \in G$ with $g(\alpha) \neq \alpha$.
(b) Show that there exists $h \in G$ with h of prime-power order such that $h(\alpha) \neq \alpha$. (That is, $h^{p^k} = 1$ for some prime p and some $k \geq 1$.)
(c) Suppose that every field F with $K \subsetneq F \subseteq L$ contains α. Show that G is cyclic of prime-power order.

6. Let A_4, the group of even permutations of 4 objects, act on $\mathbf{C}^4$ by permuting the entries of the vectors. For example, the 3-cycle $(1\,2\,4)$ maps the vector (a, b, c, d) to (d, a, c, b). This yields a representation $\rho : A_4 \to GL_4(\mathbf{C})$.
(a) (3 points) Show that ρ contains the trivial representation as a subrepresentation.
(b) (7 points) Show that ρ is the sum of two irreducible representations. (*Note:* A_4 contains the identity, eight 3-cycles, and three products of two disjoint 2-cycles.) (*Hint:* A character is irreducible if and only if its inner product with itself is 1.)